

QCMS PRINT DESIGN OPTIONS

For approval · R16 · 10-10-2026

Rule	Design A · Clean Grid	Design B · Open Lines
Background colour	None. Only section headings: light grey band	None. Only section headings: teal band, white text
Lines	Full black grid on every table	Horizontal lines only, no vertical lines
Body font	10 pt (was 5.5 - 7 pt)	10.5 pt
Row height	20 pt (about 7 mm)	22 pt (about 8 mm)
Look	Traditional QA form, easy to audit	Modern, airy, easier to read on screen

Page	Module	Design
2	Dimensional Inspection Report	Design A · Clean Grid
3	Dimensional Inspection Report	Design B · Open Lines
4	MetLAB Report	Design A · Clean Grid
5	MetLAB Report	Design B · Open Lines
6	RMTC Evaluation	Design A · Clean Grid
7	RMTC Evaluation	Design B · Open Lines
8	Bend Test Report	Design A · Clean Grid
9	Bend Test Report	Design B · Open Lines
10	OSP Inspection Report	Design A · Clean Grid
11	OSP Inspection Report	Design B · Open Lines

Reply with your choice per module, e.g. **Dimensional A, MetLAB B, RMTC A, Bend B, OSP A** — or one design for all.

 FSI FOUR STAR INDUSTRIES Pioneers in Differential Spiders	FOUR STAR INDUSTRIES PVT. LTD. DIMENSIONAL INSPECTION REPORT	Report No.	D9-DIR-2026-00112
		Format	FSI/QA/F-12
		Rev / Date	01 / 10-10-2026

REPORT DETAILS			
Part Number	40256626	Part Name	Yoke Forging
FSI Part No.	FSI-7237	Drawing / Rev	D-40256626 / C
Customer	Dana India Pvt. Ltd.	Supplier	Bharat Forge Ltd.
Heat Number	H-24578	Batch / Lot Qty	B-1167 / 1,200 pcs
Inspection Date	07-10-2026	Sample Size	5 pcs

INSPECTION RESULTS										
Sr	Characteristic	Specification	Min	Max	1	2	3	4	5	Result
1	Overall Length	120.00 ± 0.20	119.80	120.20	120.05	120.02	119.98	120.06	120.01	OK
2	Bore Diameter	Ø 35.00 +0.03/0	35.00	35.03	35.01	35.02	35.01	35.02	35.01	OK
3	Flange Thickness	12.00 ± 0.10	11.90	12.10	12.03	11.98	12.04	12.01	11.99	OK
4	Hole PCD	Ø 80.00 ± 0.05	79.95	80.05	80.01	80.02	79.99	80.00	80.03	OK
5	Bend / Flatness	0.10 max	0.00	0.10	0.04	0.05	0.03	0.06	0.04	OK
6	Hole ID	GO / NO-GO gauge	-	-	OK	OK	OK	OK	OK	OK
7	Batch Code Marking	Visual	-	-	OK	OK	OK	OK	OK	OK

DECISION			
Accepted Qty	1,200 pcs	Rejected Qty	0
Decision	ACCEPTED	Remarks	All characteristics within limits

Prepared By	Validated By	Approved By
Akshay Bharambe · 07-10-2026	Nitin Nanavare · 07-10-2026	Nitin Nanavare · 08-10-2026



FOUR STAR INDUSTRIES PVT. LTD. DIMENSIONAL INSPECTION REPORT

Report No.	D9-DIR-2026-00112
Format	FSI/QA/F-12
Rev / Date	01 / 10-10-2026

REPORT DETAILS

Part Number	40256626	Part Name	Yoke Forging
FSI Part No.	FSI-7237	Drawing / Rev	D-40256626 / C
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Heat Number	H-24578	Batch / Lot Qty	B-1167 / 1,200 pcs
Inspection Date	07-10-2026	Sample Size	5 pcs

INSPECTION RESULTS

Sr	Characteristic	Specification	Min	Max	1	2	3	4	5	Result
1	Overall Length	120.00 ± 0.20	119.80	120.20	120.05	120.02	119.98	120.06	120.01	OK
2	Bore Diameter	Ø 35.00 +0.03/0	35.00	35.03	35.01	35.02	35.01	35.02	35.01	OK
3	Flange Thickness	12.00 ± 0.10	11.90	12.10	12.03	11.98	12.04	12.01	11.99	OK
4	Hole PCD	Ø 80.00 ± 0.05	79.95	80.05	80.01	80.02	79.99	80.00	80.03	OK
5	Bend / Flatness	0.10 max	0.00	0.10	0.04	0.05	0.03	0.06	0.04	OK
6	Hole ID	GO / NO-GO gauge	-	-	OK	OK	OK	OK	OK	OK
7	Batch Code Marking	Visual	-	-	OK	OK	OK	OK	OK	OK

DECISION

Accepted Qty	1,200 pcs	Rejected Qty	0
Decision	ACCEPTED	Remarks	All characteristics within limits

Prepared By

Akshay Bharambe · 07-10-2026

Validated By

Nitin Nanavare · 07-10-2026

Approved By

Nitin Nanavare · 08-10-2026

 FSI FOUR STAR INDUSTRIES Pioneers in Differential Spiders	FOUR STAR INDUSTRIES PVT. LTD. METALLURGICAL LABORATORY REPORT	Report No.	MLAB-D9-26-00145
		Format	FSI/QA/F-21
		Rev / Date	01 / 10-10-2026

MATERIAL DETAILS

Part Number	40256626	Part Name	Yoke Forging
Material Grade	20MnCr5 (EN 10084)	Heat Number	H-24578
Supplier	Bharat Forge Ltd.	Steel Mill	JSW Steel
Heat Treatment	Case Carburising	Test Date	07-10-2026

CHEMICAL COMPOSITION

Element	Specification	Actual	Result
C %	0.17 - 0.22	0.19	OK
Mn %	1.10 - 1.40	1.24	OK
Si %	0.15 - 0.40	0.26	OK
Cr %	1.00 - 1.30	1.12	OK
S % / P %	0.035 / 0.025 max	0.021 / 0.014	OK

MECHANICAL & METALLURGICAL PROPERTIES

Parameter	Specification	Observed	Result
Surface Hardness	58 - 62 HRC	60 / 61 / 60	OK
Core Hardness	30 - 42 HRC	36	OK
Effective Case Depth @ 550 HV	0.80 - 1.10 mm	0.95 mm	OK
Grain Size (ASTM E112)	5 or finer	7	OK
Microstructure	Tempered martensite	Tempered martensite + fine carbides	OK

MICROSTRUCTURE PHOTOGRAPHS

[photo]	[photo]	[photo]
Case 500X	Core 500X	Case depth traverse

RESULT: ACCEPTED

Tested By	Verified By	Approved By
Lab Technician	Lab Incharge	QA Head



FOUR STAR INDUSTRIES PVT. LTD. METALLURGICAL LABORATORY REPORT

Report No.	MLAB-D9-26-00145
Format	FSI/QA/F-21
Rev / Date	01 / 10-10-2026

MATERIAL DETAILS

Part Number	40256626	Part Name	Yoke Forging
Material Grade	20MnCr5 (EN 10084)	Heat Number	H-24578
Supplier	Bharat Forge Ltd.	Steel Mill	JSW Steel
Heat Treatment	Case Carburising	Test Date	07-10-2026

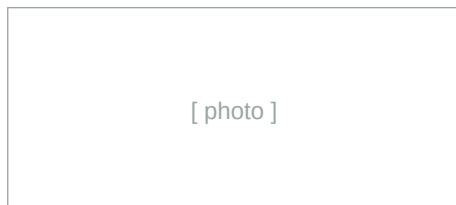
CHEMICAL COMPOSITION

Element	Specification	Actual	Result
C %	0.17 - 0.22	0.19	OK
Mn %	1.10 - 1.40	1.24	OK
Si %	0.15 - 0.40	0.26	OK
Cr %	1.00 - 1.30	1.12	OK
S % / P %	0.035 / 0.025 max	0.021 / 0.014	OK

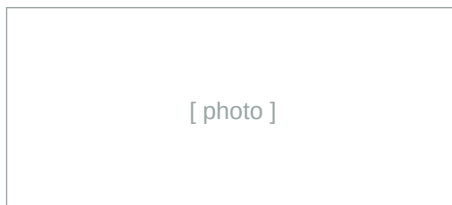
MECHANICAL & METALLURGICAL PROPERTIES

Parameter	Specification	Observed	Result
Surface Hardness	58 - 62 HRC	60 / 61 / 60	OK
Core Hardness	30 - 42 HRC	36	OK
Effective Case Depth @ 550 HV	0.80 - 1.10 mm	0.95 mm	OK
Grain Size (ASTM E112)	5 or finer	7	OK
Microstructure	Tempered martensite	Tempered martensite + fine carbides	OK

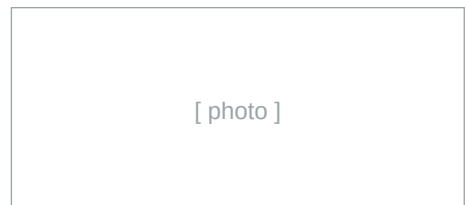
MICROSTRUCTURE PHOTOGRAPHS



Case 500X



Core 500X



Case depth traverse

RESULT: ACCEPTED

Tested By
Lab Technician

Verified By
Lab Incharge

Approved By
QA Head

 FSI FOUR STAR INDUSTRIES Pioneers in Differential Spiders	FOUR STAR INDUSTRIES PVT. LTD. RAW MATERIAL TEST CERTIFICATE EVALUATION	Report No.	RMTC-2026-00342
		Format	FSI/QA/F-05
		Rev / Date	01 / 10-10-2026

CERTIFICATE DETAILS			
Supplier	JSW Steel Ltd.	Mill TC No.	JSW/TC/88412
Heat Number	H-24578	Grade	20MnCr5
Section / Size	RCS 120 mm	Quantity	18.450 MT
Invoice / Date	INV-5521 / 02-10-2026	Applicable Parts	40256626, 40256630

CHEMICAL ANALYSIS - MILL VS SPECIFICATION					
Element	Min	Max	Mill TC	FSI Check	Result
C %	0.17	0.22	0.19	0.19	OK
Mn %	1.10	1.40	1.24	1.23	OK
Si %	0.15	0.40	0.26	0.27	OK
Cr %	1.00	1.30	1.12	1.13	OK
S %	-	0.035	0.021	0.020	OK
P %	-	0.025	0.014	0.015	OK
Al %	0.020	0.050	0.031	0.030	OK

PHYSICAL / CLEANLINESS			
Test	Specification	Actual	Result
Hardness (Annealed)	≤ 207 HB	187 HB	OK
Grain Size	5 or finer	6	OK
Inclusion Rating	As per IS 4163	A1 B1 C0 D1	OK
Jominy J9	34 - 42 HRC	38	OK
Ultrasonic Test	SEP 1921 C/c	Passed	OK

DECISION			
Decision	ACCEPTED	Released Qty	18.450 MT
Remarks	Heat released for forging	Valid For	All listed parts

Prepared By	Verified By	Approved By
QA Engineer	QA Manager	Plant Head



FOUR STAR INDUSTRIES PVT. LTD.
RAW MATERIAL TEST
CERTIFICATE EVALUATION

Report No.	RMTC-2026-00342
Format	FSI/QA/F-05
Rev / Date	01 / 10-10-2026

CERTIFICATE DETAILS

Supplier	JSW Steel Ltd.	Mill TC No.	JSW/TC/88412
Heat Number	H-24578	Grade	20MnCr5
Section / Size	RCS 120 mm	Quantity	18.450 MT
Invoice / Date	INV-5521 / 02-10-2026	Applicable Parts	40256626, 40256630

CHEMICAL ANALYSIS - MILL VS SPECIFICATION

Element	Min	Max	Mill TC	FSI Check	Result
C %	0.17	0.22	0.19	0.19	OK
Mn %	1.10	1.40	1.24	1.23	OK
Si %	0.15	0.40	0.26	0.27	OK
Cr %	1.00	1.30	1.12	1.13	OK
S %	-	0.035	0.021	0.020	OK
P %	-	0.025	0.014	0.015	OK
Al %	0.020	0.050	0.031	0.030	OK

PHYSICAL / CLEANLINESS

Test	Specification	Actual	Result
Hardness (Annealed)	≤ 207 HB	187 HB	OK
Grain Size	5 or finer	6	OK
Inclusion Rating	As per IS 4163	A1 B1 C0 D1	OK
Jominy J9	34 - 42 HRC	38	OK
Ultrasonic Test	SEP 1921 C/c	Passed	OK

DECISION

Decision	ACCEPTED	Released Qty	18.450 MT
Remarks	Heat released for forging	Valid For	All listed parts

Prepared By
QA Engineer

Verified By
QA Manager

Approved By
Plant Head

 FSI FOUR STAR INDUSTRIES Pioneers in Differential Spiders	FOUR STAR INDUSTRIES PVT. LTD. BEND TEST REPORT	Report No.	1006/2026/7237
		Format	FSI/QA/F-34
		Rev / Date	01 / 10-10-2026

REPORT DETAILS

Part #	D9OV-1167	Part Name	U-Bolt
Customer	KO	Material Used	SAE 1541
Baking Batch No.	BK-0925	Batch Quantity	2,400 pcs
Steel Heat Code	H-24578	Part Diameter	Ø 16 mm
FSI Batch No.	FSI-B-7237	HT Batch No.	HT-1185

BAKING / AGEING

Parameter	Specification	Actual	Result
Baking Temperature	400 °C / as approved process	405 °C	OK
Baking Time	60 minutes minimum	75 min	OK
Hardness	45 - 52 HRC (avg of 3)	48 / 49 / 48	OK

BEND TEST RESULTS

Test	Requirement	Observed	Status
Load Vs Cross Head Travel	Record	38.6 kN @ 42 mm	Passed
Included Bend Angle	Record	171°	Passed
Plating Surface	No peeling / flaking	No peeling, no flaking	Passed

BEND TEST PHOTOGRAPHS

[photo]			[photo]			[photo]		
Load Vs CHT graph			Bent part			Plating close-up		

Conclusion

Part bent to 171° included angle without crack, peeling or flaking. Batch ACCEPTED.

Prepared By		Verified & Approved By	
Metallurgist		QA Head	



FOUR STAR INDUSTRIES PVT. LTD. BEND TEST REPORT

Report No.	1006/2026/7237
Format	FSI/QA/F-34
Rev / Date	01 / 10-10-2026

REPORT DETAILS

Part #	D9OV-1167	Part Name	U-Bolt
Customer	KO	Material Used	SAE 1541
Baking Batch No.	BK-0925	Batch Quantity	2,400 pcs
Steel Heat Code	H-24578	Part Diameter	Ø 16 mm
FSI Batch No.	FSI-B-7237	HT Batch No.	HT-1185

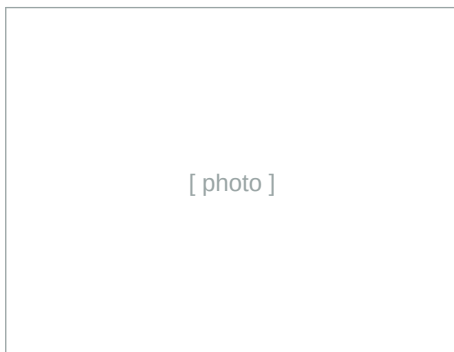
BAKING / AGEING

Parameter	Specification	Actual	Result
Baking Temperature	400 °C / as approved process	405 °C	OK
Baking Time	60 minutes minimum	75 min	OK
Hardness	45 - 52 HRC (avg of 3)	48 / 49 / 48	OK

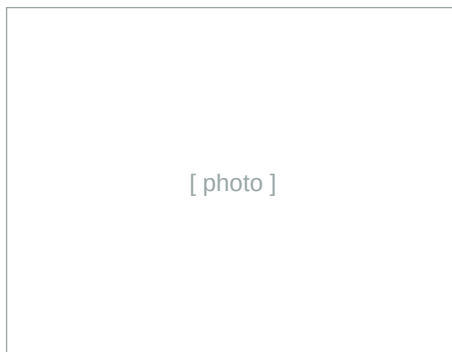
BEND TEST RESULTS

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Load Vs Cross Head Travel	Record	38.6 kN @ 42 mm	Passed
Included Bend Angle	Record	171°	Passed
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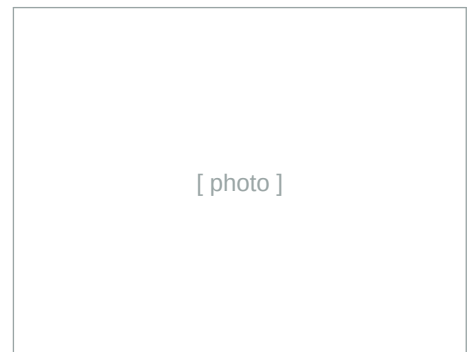
BEND TEST PHOTOGRAPHS



Load Vs CHT graph



Bent part



Plating close-up

Conclusion Part bent to 171° included angle without crack, peeling or flaking. Batch ACCEPTED.

Prepared By
Metallurgist

Verified & Approved By
QA Head

 FSI FOUR STAR INDUSTRIES Pioneers in Differential Spiders	FOUR STAR INDUSTRIES PVT. LTD. OSP INSPECTION REPORT · POST-RECEIPT	Report No.	D9-OSP-2026-00058
		Format	FSI/QA/F-41
		Rev / Date	01 / 10-10-2026

OSP BATCH DETAILS

OSP Job No.	OSP-D9-2026-00058	Process	Zinc Flake Coating
Vendor	Surface Tech Pvt. Ltd.	Vendor Batch	ST-9921
Part Number	40256626	FSI Batch	FSI-B-7237
Heat Number	H-24578	Qty Received	1,200 pcs

PROCESS-SPECIFIC PARAMETERS

Sr	Parameter	Specification	Min	Max	Actual	Result
1	Coating Thickness	8 - 12 µm	8	12	10.2	OK
2	Adhesion (Cross-cut)	GT0 / GT1	-	-	GT0	OK
3	Salt Spray	720 h no red rust	-	-	Passed (lab)	OK
4	Appearance	Uniform silver, no bare spots	-	-	OK	OK
5	Thread Gauge	GO / NO-GO	-	-	OK	OK

OSP QUALITY GATE

Gate Report	Status
Sample Dimensional	Finalized · Accepted
Sample MetLAB	Finalized · Accepted
Receipt Dimensional	Finalized · Accepted
Receipt MetLAB	Finalized · Accepted

OSP BATCH RELEASED

Prepared By	Validated By	Approved By
Inspector	QA Engineer	QA Manager



FOUR STAR INDUSTRIES PVT. LTD.

OSP INSPECTION REPORT ·

POST-RECEIPT

Report No.	D9-OSP-2026-00058
Format	FSI/QA/F-41
Rev / Date	01 / 10-10-2026

OSP BATCH DETAILS

OSP Job No.	OSP-D9-2026-00058	Process	Zinc Flake Coating
Vendor	Surface Tech Pvt. Ltd.	Vendor Batch	ST-9921
Part Number	40256626	FSI Batch	FSI-B-7237
Heat Number	H-24578	Qty Received	1,200 pcs

PROCESS-SPECIFIC PARAMETERS

Sr	Parameter	Specification	Min	Max	Actual	Result
1	Coating Thickness	8 - 12 µm	8	12	10.2	OK
2	Adhesion (Cross-cut)	GT0 / GT1	-	-	GT0	OK
3	Salt Spray	720 h no red rust	-	-	Passed (lab)	OK
4	Appearance	Uniform silver, no bare spots	-	-	OK	OK
5	Thread Gauge	GO / NO-GO	-	-	OK	OK

OSP QUALITY GATE

Gate Report	Status
Sample Dimensional	Finalized · Accepted
Sample MetLAB	Finalized · Accepted
Receipt Dimensional	Finalized · Accepted
Receipt MetLAB	Finalized · Accepted

OSP BATCH RELEASED

Prepared By
Inspector

Validated By
QA Engineer

Approved By
QA Manager